GE Healthcare

Signa® Phased Array Shoulder Coil

Service Manual

Direction 2135786
Rev 2

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REVISION HISTORY

REV	DATE	AUTHOR	REASON FOR CHANGE
0	Sept. 9, 1998	L. Loehrer	Converted Toolbook document to MS Word 7.0.
1	May 20, 1999	S.M. Atladottir	Updated procedure references for new GUI.
2	May 28, 2010	R. David	Removed PIN Diode Replacement section PIN Diode FRU.

DAMAGE IN TRANSPORTATION

All packages should be closely examined at time of delivery. If damage is apparent, have notation "damage in shipment" written on all copies of the freight or express bill before delivery is accepted or "signed for" by a General Electric representative or a hospital receiving agent. Whether noted or concealed, damage MUST be reported to the carrier immediately upon discovery, or in any event, within 14 days after receipt, and the contents and containers held for inspection by the carrier. A transportation company will not pay a claim for damage if an inspection is not requested within this 14-day period.

To file a report:

Call 1-800-548-3366 and use option 6.

Contact your local service coordinator for more information on this process.

LANGUAGE POLICY

ПРЕДУПРЕЖДЕНИЕ (BG)

ТОВА УПЪТВАНЕ ЗА РАБОТА Е НАЛИЧНО САМО НА АНГЛИЙСКИ ЕЗИК.

АКО ДОСТАВЧИКЪТ НА УСЛУГАТА НА КЛИЕНТА ИЗИСКА ЕЗИК, РАЗЛИЧЕН ОТ АНГЛИЙСКИ, ЗАДЪЛЖЕНИЕ НА КЛИЕНТА Е ДА ОСИГУРИ ПРЕВОД.

- НЕ ИЗПОЛЗВАЙТЕ ОБОРУДВАНЕТО ПРЕДИ ДА СТЕ СЕ КОНСУЛТИРАЛИ И РАЗБРАЛИ УПЪТВАНЕТО ЗА РАБОТА.
- НЕСПАЗВАНЕТО НА ТОВА ПРЕДУПРЕЖДЕНИЕ МОЖЕ ДА ДОВЕДЕ ДО НАРАНЯВАНЕ НА ДОСТАВЧИКА НА УСЛУГАТА, ОПЕРАТОРА ИЛИ ПАЦИЕНТ В РЕЗУЛТАТ НА ТОКОВ УДАР ИЛИ МЕХАНИЧНА ИЛИ ДРУГА ОПАСНОСТ.

警告 (ZH-CN)

本维修手册仅提供英文版本。

- 如果维修服务提供商需要非英文版本，客户需自行提供翻译服务。
- 未详细阅读和完全理解本维修手册之前，不得进行维修。
- 忽略本警告可能对维修人员，操作员或患者造成触电、机械伤害或其他形式的伤害。

警告 (ZN-HK)

本服務手冊僅提供英文版本。

- 倘若客戶的服務供應商需要英文以外之服務手冊，客戶有責任提供翻譯服務。
- 除非已參閱本服務手冊及明白其內容，否則切勿嘗試維修設備。
- 不遵從本警告或會令服務供應商、網絡供應商或病人受到觸電、機械性或其他的危險。

警告 (ZH-TW)

本維修手冊僅有英文版。

- 若客戶的維修廠商需要英文版以外的語言，應由客戶自行提供翻譯服務。
- 請勿試圖維修本設備，除非 您已查閱並瞭解本維修手冊。
- 若未留意本警告，可能導致維修廠商、操作員或病患因觸電、機械或其他危險而受傷。

UPOZORENJE (HR)

OVAJ SERVISNI PRIRUČNIK DOSTUPAN JE NA ENGLESKOM JEZIKU.

- AKO DAVATELJ USLUGE KLIJENTA TREBA NEKI DRUGI JEZIK, KLIJENT JE DUŽAN OSIGURATI PRIJEVOD.
- NE POKUŠAVAJTE SERVISIRATI OPREMU AKO NISTE U POTPUNOSTI PROČITALI I RAZUMJELI OVAJ SERVISNI PRIRUČNIK.
- ZANEMARITE LI OVO UPOZORENJE, MOŽE DOĆI DO OZLJEDE DAVATELJA USLUGE, OPERATERA ILI PACIJENTA USLIJED STRUJNOG UDARA, MEHANIČKIH ILI DRUGIH RIZIKA.

VÝSTRAHA

(CS)

TENTO PROVOZNÍ NÁVOD EXISTUJE POUZE V ANGLICKÉM JAZYCE.

- PŘÍPADĚ, ŽE EXTERNÍ SLUŽBA ZÁKAZNÍKŮM POTŘEBUJE NÁVOD V JINÉM JAZYCE, JE ZAJIŠTĚNÍ PŘEKLADU DO ODPOVÍDAJÍCÍHO JAZYKA ÚKOLEM ZÁKAZNÍKA.
- NESNAŽTE SE O ÚDRŽBU TOHOTO ZAŘÍZENÍ, ANIŽ BYSTE SI PŘEČETLI TENTO PROVOZNÍ NÁVOD A POCHOPILI JEHO OBSAH.
- PŘÍPADĚ NEDODRŽOVÁNÍ TÉTO VÝSTRAHY MŮŽE DOJÍT K PORANĚNÍ PRACOVNÍKA PRODEJNÍHO SERVISU, OBSLUŽNÉHO PERSONÁLU NEBO PACIENTŮ VLIVEM ELEKTRICKÉHO PROUDU, RESPEKTIVE VLIVEM MECHANICKÝCH ČI JINÝCH RIZIK.

ADVARSEL

(DA)

DENNE SERVICEMANUAL FINDES KUN PÅ ENGELSK.

- HVIS EN KUNDES TEKNIKER HAR BRUG FOR ET ANDET SPROG END ENGELSK, ER DET KUNDENS ANSVAR AT SØRGE FOR OVERSÆTTELSE.
- FORSØG IKKE AT SERVICERE UDSTYRET MEDMINDRE DENNE SERVICEMANUAL HAR VÆRET KONSULTERET OG ER FORSTÅET.
- MANGLENDE OVERHOLDELSE AF DENNE ADVARSEL KAN MEDFØRE SKADE PÅ GRUND AF ELEKTRISK, MEKANISK ELLER ANDEN FARE FOR TEKNIKEREN, OPERATØREN ELLER PATIENTEN.

WAARSCHUWING

(NL)

DEZE ONDERHOUDSHANDLEIDING IS ENKEL IN HET ENGELS VERKRIJGBAAR.

- ALS HET ONDERHOUDSPERSONEEL EEN ANDERE TAAL VEREIST, DAN IS DE KLANT VERANTWOORDELIJK VOOR DE VERTALING ERVAN.
- PROBEER DE APPARATUUR NIET TE ONDERHOUDEN VOORDAT DEZE ONDERHOUDSHANDLEIDING WERD GERAADPLEEGD EN BEGREPEN IS.
- INDIEN DEZE WAARSCHUWING NIET WORDT OPGEVOLGD, ZOU HET ONDERHOUDSPERSONEEL, DE OPERATOR OF EEN PATIËNT GEWOND KUNNEN RAKEN ALS GEVOLG VAN EEN ELEKTRISCHE SCHOK, MECHANISCHE OF ANDERE GEVAREN.

ENGLISH

(EN)

THIS SERVICE MANUAL IS AVAILABLE IN ENGLISH ONLY.

- IF A CUSTOMER'S SERVICE PROVIDER REQUIRES A LANGUAGE OTHER THAN ENGLISH, IT IS THE CUSTOMER'S RESPONSIBILITY TO PROVIDE TRANSLATION SERVICES.
- DO NOT ATTEMPT TO SERVICE THE EQUIPMENT UNLESS THIS SERVICE MANUAL HAS BEEN CONSULTED AND IS UNDERSTOOD.
- FAILURE TO HEED THIS WARNING MAY RESULT IN INJURY TO THE SERVICE PROVIDER, OPERATOR OR PATIENT FROM ELECTRIC SHOCK, MECHANICAL OR OTHER HAZARDS.

HOIATUS
(ET)

SEE TEENINDUSJUHEND ON SAADAVAL AINULT INGLISE KEELES.

- KUI KLIENDITEENINDUSE OSUTAJA NÕUAB JUHENDIT INGLISE KEELEST ERINEVAS KEELES, VASTUTAB KLIENT TÕLKETEENUSE OSUTAMISE EEST.
- ÄRGE ÜRITAGE SEADMEID TEENINDADA ENNE EELNEVALT KÄESOLEVA TEENINDUSJUHENDIGA TUTVUMIST JA SELLEST ARU SAAMIST.
- KÄESOLEVA HOIATUSE EIRAMINE VÕIB PÕHJUSTADA TEENUSEOSUTAJA, OPERAATORI VÕI PATSIENDI VIGASTAMIST ELEKTRILÖÖGI, MEHAANILISE VÕI MUU OHU TAGAJÄRJEL.

VAROITUS
(FI)

TÄMÄ HUOLTO-OHJE ON SAATAVILLA VAIN ENGLANNIKSI.

- JOS ASIAKKAAN HUOLTOHENKILÖSTÖ VAATII MUUTA KUIN ENGLANNINKIELISTÄ MATERIAALIA, TARVITTAVAN KÄÄNNÖKSEN HANKKIMINEN ON ASIAKKAAN VASTUULLA.
- ÄLÄ YRITÄ KORJATA LAITTEISTOA ENNEN KUIN OLET VARMASTI LUKENUT JA YMMÄRTÄNYT TÄMÄN HUOLTO-OHJEEN.
- MIKÄLI TÄTÄ VAROITUSTA EI NOUDATETA, SEURAUKSENA VOI OLLA HUOLTOHENKILÖSTÖN, LAITTEISTON KÄYTTÄJÄN TAI POTILAAN VAHINGOITTUMINEN SÄHKÖISKUN, MEKAANISEN VIAN TAI MUUN VAARATILANTEEN VUOKSI.

ATTENTION
(FR)

CE MANUEL DE SERVICE N'EST DISPONIBLE QU'EN ANGLAIS.

- SI LE TECHNICIEN DU CLIENT A BESOIN DE CE MANUEL DANS UNE AUTRE LANGUE QUE L'ANGLAIS, C'EST AU CLIENT QU'IL INCOMBE DE LE FAIRE TRADUIRE.
- NE PAS TENTER D'INTERVENIR SUR LES EQUIPEMENTS TANT QUE LE MANUEL SERVICE N'A PAS ETE CONSULTE ET COMPRIS
- LE NON-RESPECT DE CET AVERTISSEMENT PEUT ENTRAÎNER CHEZ LE TECHNICIEN, L'OPÉRATEUR OU LE PATIENT DES BLESSURES DUES À DES DANGERS ÉLECTRIQUES, MÉCANIQUES OU AUTRES.

WARNUNG
(DE)

DIESE SERVICEANLEITUNG EXISTIERT NUR IN ENGLISCHER SPRACHE.

- FALLS EIN FREMDER KUNDENDIENST EINE ANDERE SPRACHE BENÖTIGT, IST ES AUFGABE DES KUNDEN FÜR EINE ENTSPRECHENDE ÜBERSETZUNG ZU SORGEN.
- VERSUCHEN SIE NICHT DIESE ANLAGE ZU WARTEN, OHNE DIESE SERVICEANLEITUNG GELESEN UND VERSTANDEN ZU HABEN.
- WIRD DIESE WARNUNG NICHT BEACHTET, SO KANN ES ZU VERLETZUNGEN DES KUNDENDIENSTTECHNIKERS, DES BEDIENERS ODER DES PATIENTEN DURCH STROMSCHLÄGE, MECHANISCHE ODER SONSTIGE GEFAHREN KOMMEN.

ΠΡΟΕΙΔΟΠΟΙΗΣΗ
(EL)

ΤΟ ΠΑΡΟΝ ΕΓΧΕΙΡΙΔΙΟ ΣΕΡΒΙΣ ΔΙΑΤΙΘΕΤΑΙ ΣΤΑ ΑΓΓΛΙΚΑ ΜΟΝΟ.

- ΕΑΝ ΤΟ ΑΤΟΜΟ ΠΑΡΟΧΗΣ ΣΕΡΒΙΣ ΕΝΟΣ ΠΕΛΑΤΗ ΑΠΑΙΤΕΙ ΤΟ ΠΑΡΟΝ ΕΓΧΕΙΡΙΔΙΟ ΣΕ ΓΛΩΣΣΑ ΕΚΤΟΣ ΤΩΝ ΑΓΓΛΙΚΩΝ, ΑΠΟΤΕΛΕΙ ΕΥΘΥΝΗ ΤΟΥ ΠΕΛΑΤΗ ΝΑ ΠΑΡΕΧΕΙ ΥΠΗΡΕΣΙΕΣ ΜΕΤΑΦΡΑΣΗΣ.
- ΜΗΝ ΕΠΙΧΕΙΡΗΣΕΤΕ ΤΗΝ ΕΚΤΕΛΕΣΗ ΕΡΓΑΣΙΩΝ ΣΕΡΒΙΣ ΣΤΟΝ ΕΞΟΠΛΙΣΜΟ ΕΚΤΟΣ ΕΑΝ ΕΧΕΤΕ ΣΥΜΒΟΥΛΕΥΤΕΙ ΚΑΙ ΕΧΕΤΕ ΚΑΤΑΝΟΗΣΕΙ ΤΟ ΠΑΡΟΝ ΕΓΧΕΙΡΙΔΙΟ ΣΕΡΒΙΣ.
- ΕΑΝ ΔΕ ΛΑΒΕΤΕ ΥΠΟΨΗ ΤΗΝ ΠΡΟΕΙΔΟΠΟΙΗΣΗ ΑΥΤΗ, ΕΝΔΕΧΕΤΑΙ ΝΑ ΠΡΟΚΛΗΘΕΙ ΤΡΑΥΜΑΤΙΣΜΟΣ ΣΤΟ ΑΤΟΜΟ ΠΑΡΟΧΗΣ ΣΕΡΒΙΣ, ΣΤΟ ΧΕΙΡΙΣΤΗ Ή ΣΤΟΝ ΑΣΘΕΝΗ ΑΠΟ ΗΛΕΚΤΡΟΠΛΗΞΙΑ, ΜΗΧΑΝΙΚΟΥΣ Ή ΑΛΛΟΥΣ ΚΙΝΔΥΝΟΥΣ.

FIGYELMEZTETÉS
(HU)

EZEN KARBANTARTÁSI KÉZIKÖNYV KIZÁRÓLAG ANGOL NYELVEN ÉRHETŐ EL.

- HA A VEVŐ SZOLGÁLTATÓJA ANGOLTÓL ELTÉRŐ NYELVRE TART IGÉNYT, AKKOR A VEVŐ FELELŐSSÉGE A FORDÍTÁS ELKÉSZÍTTETÉSE.
- NE PRÓBÁLJA ELKEZDENI HASZNÁLNI A BERENDEZÉST, AMÍG A KARBANTARTÁSI KÉZIKÖNYVBEN LEÍRTAKAT NEM ÉRTELMEZTÉK.
- EZEN FIGYELMEZTETÉS FIGYELMEN KÍVÜL HAGYÁSA A SZOLGÁLTATÓ, MŰKÖDTETŐ VAGY A BETEG ÁRAMÜTÉS, MECHANIKAI VAGY EGYÉB VESZÉLYHELYZET MIATTI SÉRÜLÉSÉT EREDMÉNYEZHETI.

AÐVÖRUN
(IS)

ÞESSI ÞJÓNUSTUHANDBÓK ER EINGÖNGU FÁANLEG Á ENSKU.

- EF AÐ ÞJÓNUSTUVEITANDI VIÐSKIPTAMANNS ÞARFNAST ANNAS TUNGUMÁLS EN ENSKU, ER ÞAÐ SKYLDA VIÐSKIPTAMANNS AÐ SKAFFA TUNGUMÁLAÞJÓNUSTU.
- REYNIÐ EKKI AÐ AFGREIÐA TÆKIÐ NEMA AÐ ÞESSI ÞJÓNUSTUHANDBÓK
HEFUR VERIÐ SKOÐUÐ OG SKILIN.
- BROT Á SINNA ÞESSARI AÐVÖRUN GETUR LEITT TIL MEIÐSLA Á ÞJÓNUSTUVEITANDA, STJÓRNANDA EÐA SJÚKLINGS FRÁ RAFLOSTI, VÉLRÆNU EÐA ÖÐRUM ÁHÆTTUM.

AVVERTENZA
(IT)

IL PRESENTE MANUALE DI MANUTENZIONE E DISPONIBILE SOLTANTO IN INGLESE.

- SE UN ADDETTO ALLA MANUTENZIONE ESTERNO ALLA GEMS RICHIEDE IL MANUALE IN UNA LINGUA DIVERSA, IL CLIENTE E TENUTO A PROVVEDERE DIRETTAMENTE ALLA TRADUZIONE.
- SI PROCEDA ALLA MANUTENZIONE DELL'APPARECCHIATURA SOLO DOPO AVER
CONSULTATO IL PRESENTE MANUALE ED AVERNE COMPRESO IL CONTENUTO
- IL NON RISPETTO DELLA PRESENTE AVVERTENZA POTREBBE FAR COMPIERE OPERAZIONI DA CUI DERIVINO LESIONI ALL'ADDETTO ALLA MANUTENZIONE, ALL'UTILIZZATORE ED AL PAZIENTE PER FOLGORAZIONE ELETTRICA, PER URTI MECCANICI OD ALTRI RISCHI.

警告

(JA)

このサービスマニュアルには英語版しかありません。

- サービスを担当される業者が英語以外の言語を要求される場合、翻訳作業はその業者の責任で行うものとさせていただきます。
- このサービスマニュアルを熟読し理解せずに、装置のサービスを行わないでください。
- この警告に従わない場合、サービスを担当される方、操作員あるいは患者さんが、感電や機械的又はその他の危険により負傷する可能性があります。

경고

(KO)

본 서비스 매뉴얼은 영어로만 이용하실 수 있습니다.

- 고객의 서비스 제공자가 영어 이외의 언어를 요구할 경우, 번역 서비스를 제공하는 것은 고객의 책임입니다.
- 본 서비스 지침서를 참고했고 이해하지 않는 한은 해당 장비를 수리하려고 시도하지 마십시오.
- 이 경고에 유의하지 않으면 전기 쇼크, 기계상의 혹은 다른 위험으로부터 서비스 제공자, 운영자 혹은 환자에게 위해를 가할 수 있습니다.

BRĪDINĀJUMS

(LV)

ŠĪ APKALPES ROKASGRĀMATA IR PIEEJAMA TIKAI ANGĻU VALODĀ.

- JA KLIENTA APKALPES SNIEDZĒJAM NEPIECIEŠAMA INFORMĀCIJA CITĀ VALODĀ, NEVIS ANGĻU, KLIENTA PIENĀKUMS IR NODROŠINĀT TULKOŠANU.
- NEVEICIET APRĪKOJUMA APKALPI BEZ APKALPES ROKASGRĀMATAS IZLASĪŠANAS UN SAPRAŠANAS.
- ŠĪ BRĪDINĀJUMA NEIEVĒROŠANA VAR RADĪT ELEKTRISKĀS STRĀVAS TRIECIENA, MEHĀNISKU VAI CITU RISKU IZRAISĪTU TRAUMU APKALPES SNIEDZĒJAM, OPERATORAM VAI PACIENTAM.

ĮSPĖJIMAS

(LT)

ŠIS EKSPLOATAVIMO VADOVAS YRA PRIEINAMAS TIK ANGLŲ KALBA.

- JEI KLIENTO PASLAUGŲ TIEKĖJAS REIKALAUJA VADOVO KITA KALBA – NE ANGLŲ, NUMATYTI VERTIMO PASLAUGAS YRA KLIENTO ATSAKOMYBĖ.
- NEMĖGINKITE ATLIKTI ĮRANGOS TECHNINĖS PRIEŽIŪROS, NEBENT ATSIŽVELGĖTE Į ŠĮ EKSPLOATAVIMO VADOVĄ IR JĮ SUPRATOTE.
- JEI NEATKREIPSITE DĖMESIO Į ŠĮ PERSPĖJIMĄ, GALIMI SUŽALOJIMAI DĖL ELEKTROS ŠOKO,
- MECHANINIŲ AR KITŲ PAVOJŲ PASLAUGŲ TIEKĖJUI, OPERATORIUI AR PACIENTUI.

ADVARSEL

(NO)

DENNE SERVICEHÅNDBOKEN FINNES BARE PÅ ENGELSK.

- HVIS KUNDENS SERVICELEVERANDØR TRENGER ET ANNET SPRÅK, ER DET KUNDENS ANSVAR Å SØRGE FOR OVERSETTELSE.
- IKKE FORSØK Å REPARERE UTSTYRET UTEN AT DENNE SERVICEHÅNDBOKEN ER LEST OG FORSTÅTT.
- MANGLENDE HENSYN TIL DENNE ADVARSELEN KAN FØRE TIL AT SERVICELEVERANDØREN, OPERATØREN ELLER PASIENTEN SKADES PÅ GRUNN AV ELEKTRISK STØT, MEKANISKE ELLER ANDRE FARER.

OSTRZEŻENIE
(PL)

NINIEJSZY PODRĘCZNIK SERWISOWY DOSTĘPNY JEST JEDYNIE W JĘZYKU ANGIELSKIM.

- JEŚLI DOSTAWCA USŁUG KLIENTA WYMAGA JĘZYKA INNEGO NIŻ ANGIELSKI, ZAPEWNIENIE USŁUGI TŁUMACZENIA JEST OBOWIĄZKIEM KLIENTA.
- NIE PRÓBOWAĆ SERWISOWAĆ WYPOSAŻENIA BEZ ZAPOZNANIA SIĘ I ZROZUMIENIA NINIEJSZEGO PODRĘCZNIKA SERWISOWEGO.
- NIEZASTOSOWANIE SIĘ DO TEGO OSTRZEŻENIA MOŻE SPOWODOWAĆ URAZY DOSTAWCY USŁUG, OPERATORA LUB PACJENTA W WYNIKU PORAŻENIA ELEKTRYCZNEGO, ZAGROŻENIA MECHANICZNEGO BĄDŹ INNEGO.

AVISO
(PT-BR)

ESTE MANUAL DE ASSISTÊNCIA TÉCNICA SÓ SE ENCONTRA DISPONÍVEL EM INGLÊS.

- SE QUALQUER OUTRO SERVIÇO DE ASSISTÊNCIA TÉCNICA, QUE NÃO A GEMS, SOLICITAR ESTES MANUAIS NOUTRO IDIOMA, É DA RESPONSABILIDADE DO CLIENTE FORNECER OS SERVIÇOS DE TRADUÇÃO.
- NÃO TENTE REPARAR O EQUIPAMENTO SEM TER CONSULTADO ECOMPREENDIDO ESTE MANUAL DE ASSISTÊNCIA TÉCNICA.

O NÃO CUMPRIMENTO DESTE AVISO PODE POR EM PERIGO A SEGURANÇA DO TÉCNICO, OPERADOR OU PACIENTE DEVIDO A CHOQUES ELÉTRICOS, MECÂNICOS OU OUTROS.

ATENÇÃO
(PT-PT)

ESTE MANUAL DE ASSISTÊNCIA TÉCNICA SÓ SE ENCONTRA DISPONÍVEL EM INGLÊS.

- SE QUALQUER OUTRO SERVIÇO DE ASSISTÊNCIA TÉCNICA, QUE NÃO A GEMS, SOLICITAR ESTES MANUAIS NOUTRO IDIOMA, É DA RESPONSABILIDADE DO CLIENTE FORNECER OS SERVIÇOS DE TRADUÇÃO.
- NÃO TENTE REPARAR O EQUIPAMENTO SEM TER CONSULTADO ECOMPREENDIDO ESTE MANUAL DE ASSISTÊNCIA TÉCNICA
- O NÃO CUMPRIMENTO DESTE AVISO PODE COLOCAR EM PERIGO A SEGURANÇA DO TÉCNICO, DO OPERADOR OU DO PACIENTE DEVIDO A CHOQUES ELÉCTRICOS, MECÂNICOS OU OUTROS.

ATENȚIE
(RO)

ACEST MANUAL DE SERVICE ESTE DISPONIBIL NUMAI ÎN LIMBA ENGLEZĂ.

- DACĂ UN FURNIZOR DE SERVICII PENTRU CLIENȚI NECESITĂ O ALTĂ LIMBĂ DECÂT CEA ENGLEZĂ, ESTE DE DATORIA CLIENTULUI SĂ FURNIZEZE O TRADUCERE.
- NU ÎNCERCAȚI SĂ REPARAȚI ECHIPAMENTUL DECÂT ULTERIOR CONSULTĂRII ŞI ÎNȚELEGERII ACESTUI MANUAL DE SERVICE.
- IGNORAREA ACESTUI AVERTISMENT AR PUTEA DUCE LA RĂNIREA DEPANATORULUI, OPERATORULUI SAU PACIENTULUI ÎN URMA PERICOLELOR DE ELECTROCUTARE, MECANICE SAU DE ALTĂ NATURĂ.

ОСТОРОЖНО!
(RU)

ДАННОЕ РУКОВОДСТВО ПО ОБСЛУЖИВАНИЮ ПРЕДЛАГАЕТСЯ ТОЛЬКО НА АНГЛИЙСКОМ ЯЗЫКЕ.

- ЕСЛИ СЕРВИСНОМУ ПЕРСОНАЛУ КЛИЕНТА НЕОБХОДИМО РУКОВОДСТВО НЕ НА АНГЛИЙСКОМ, А НА КАКОМ-ТО ДРУГОМ ЯЗЫКЕ, КЛИЕНТУ СЛЕДУЕТ САМОСТОЯТЕЛЬНО ОБЕСПЕЧИТЬ ПЕРЕВОД.
- ПЕРЕД ОБСЛУЖИВАНИЕМ ОБОРУДОВАНИЯ ОБЯЗАТЕЛЬНО ОБРАТИТЕСЬ К ДАННОМУ РУКОВОДСТВУ И ПОЙМИТЕ ИЗЛОЖЕННЫЕ В НЕМ СВЕДЕНИЯ.
- НЕСОБЛЮДЕНИЕ ТРЕБОВАНИЙ ДАННОГО ПРЕДУПРЕЖДЕНИЯ МОЖЕТ ПРИВЕСТИ К ТОМУ, ЧТО СПЕЦИАЛИСТ ПО ОБСЛУЖИВАНИЮ, ОПЕРАТОР ИЛИ ПАЦИЕНТ ПОЛУЧАТ УДАР ЭЛЕКТРИЧЕСКИМ ТОКОМ, МЕХАНИЧЕСКУЮ ТРАВМУ ИЛИ ДРУГОЕ ПОВРЕЖДЕНИЕ.

UPOZORENJE
(SR)

OVO SERVISNO UPUTSTVO JE DOSTUPNO SAMO NA ENGLESKOM JEZIKU.

- AKO KLIJENTOV SERVISER ZAHTEVA NEKI DRUGI JEZIK, KLIJENT JE DUŽAN DA OBEZBEDI PREVODILAČKE USLUGE.
- NE POKUŠAVAJTE DA OPRAVITE UREĐAJ AKO NISTE PROČITALI I RAZUMELI OVO SERVISNO UPUTSTVO.
- ZANEMARIVANJE OVOG UPOZORENJA MOŽE DOVESTI DO POVREĐIVANJA SERVISERA, RUKOVAOCA ILI PACIJENTA USLED STRUJNOG UDARA ILI MEHANIČKIH I DRUGIH OPASNOSTI.

UPOZORNENIE
(SK)

TENTO NÁVOD NA OBSLUHU JE K DISPOZÍCII LEN V ANGLIČTINE.

- AK ZÁKAZNÍKOV POSKYTOVATEĽ SLUŽIEB VYŽADUJE INÝ JAZYK AKO ANGLIČTINU, POSKYTNUTIE PREKLADATEĽSKÝCH SLUŽIEB JE ZODPOVEDNOSŤOU ZÁKAZNÍKA.
- NEPOKÚŠAJTE SA O OBSLUHU ZARIADENIA SKÔR, AKO SI NEPREČÍTATE NÁVOD NA OBLUHU A NEPOROZUMIETE MU.
- ZANEDBANIE TOHTO UPOZORNENIA MÔŽE VYÚSTIŤ DO ZRANENIA POSKYTOVATEĽA SLUŽIEB, OBSLUHUJÚCEJ OSOBY ALEBO PACIENTA ELEKTRICKÝM PRÚDOM, DO MECHANICKÉHO ALEBO INÉHO NEBEZPEČENSTVA.

ATENCION
(ES)

ESTE MANUAL DE SERVICIO SOLO EXISTE EN INGLES.

- SI ALGUN PROVEEDOR DE SERVICIOS AJENO A GEMS SOLICITA UN IDIOMA QUE NO SEA EL INGLES, ES RESPONSABILIDAD DEL CLIENTE OFRECER UN SERVICIO DE TRADUCCION
- NO SE DEBERA DAR SERVICIO TECNICO AL EQUIPO, SIN HABER CONSULTADO Y COMPRENDIDO ESTE MANUAL DE SERVICIO
- LA NO OBSERVANCIA DEL PRESENTE AVISO PUEDE DAR LUGAR A QUE EL PROVEEDOR DE SERVICIOS, EL OPERADOR O EL PACIENTE SUFRAN LESIONES PROVOCADAS POR CAUSAS ELÉCTRICAS, MECÁNICAS O DE OTRA NATURALEZA.

VARNING
(SV)

DEN HÄR SERVICEHANDBOKEN FINNS BARA TILLGÄNGLIG PÅ ENGELSKA.

- OM EN KUNDS SERVICETEKNIKER HAR BEHOV AV ETT ANNAT SPRÅK ÄN ENGELSKA ANSVARAR KUNDEN FÖR ATT TILLHANDAHÅLLA ÖVERSÄTTNINGSTJÄNSTER.
- FÖRSÖK INTE UTFÖRA SERVICE PÅ UTRUSTNINGEN OM DU INTE HAR LÄST OCH FÖRSTÅR DEN HÄR SERVICEHANDBOKEN.
- OM DU INTE TAR HÄNSYN TILL DEN HÄR VARNINGEN KAN DET RESULTERA I SKADOR PÅ SERVICETEKNIKERN, OPERATÖREN ELLER PATIENTEN TILL FÖLJD AV ELEKTRISKA STÖTAR, MEKANISKA FAROR ELLER ANDRA FAROR.

OPOZORILO
(SL)

TA SERVISNI PRIROČNIK JE NA VOLJO SAMO V ANGLEŠKEM JEZIKU.

- ČE PONUDNIK STORITVE STRANKE POTREBUJE PRIROČNIK V DRUGEM JEZIKU, MORA STRANKA ZAGOTOVITI PREVOD.
- NE POSKUŠAJTE SERVISIRATI OPREME, ČE TEGA PRIROČNIKA NISTE V CELOTI PREBRALI IN RAZUMELI.
- ČE TEGA OPOZORILA NE UPOŠTEVATE, SE LAHKO ZARADI ELEKTRIČNEGA UDARA, MEHANSKIH ALI DRUGIH NEVARNOSTI POŠKODUJE PONUDNIK STORITEV, OPERATER ALI BOLNIK.

DİKKAT
(TR)

BU SERVİS KILAVUZUNUN SADECE İNGİLİZCESİ MEVCUTTUR.

- EĞER MÜŞTERİ TEKNİSYENİ BU KILAVUZU İNGİLİZCE DIŞINDA BİR BAŞKA LİSANDAN TALEP EDERSE, BUNU TERCÜME ETTİRMEK MÜŞTERİYE DÜŞER.
- SERVİS KILAVUZUNU OKUYUP ANLAMADAN EKİPMANLARA MÜDAHALE ETMEYİNİZ.
- BU UYARIYA UYULMAMASI, ELEKTRİK, MEKANİK VEYA DİĞER TEHLİKELERDEN DOLAYI TEKNİSYEN, OPERATÖR VEYA HASTANIN YARALANMASINA YOL AÇABİLİR.

SECTION 1 - INTRODUCTION

1-1 Phased Array Shoulder Coil Operation

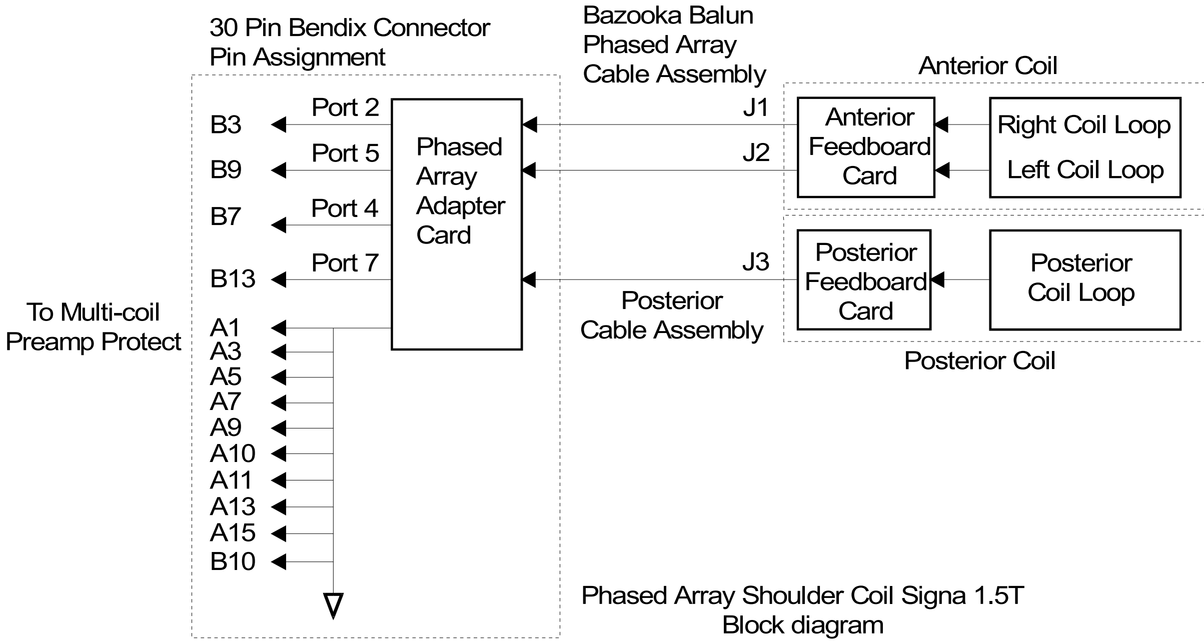
The Signa Phased Array Shoulder Coil is designed for Shoulder imaging on Signa 4.X and 5.X platforms. The coil operates as a two or three element array coil on Signa systems with Phased Array capability. It is a receive-only coil in both modes. Each of the two coil elements in the anterior unit drives a separate Phased Array coil port, the posterior unit drives a third Phased Array coil port.

The anterior unit consists of two single loop elements, the two elements are critically overlapped to minimize coupling between them. Each loop has identical RF circuitry, and each loop drives a separate output cable. Isolation of signal for the two outputs from each loop is accomplished by positioning the overlap of the two loops so as to cancel the coupling between them. The posterior unit consists of a single-loop element. Isolation of the posterior loop from the anterior loop pair is accomplished by physical spacing of the coil elements and the preamplifier decoupling concept utilized with other Phased Array coils. The posterior unit plugs into a connector on the anterior unit connector plug, allowing it to be used or not at the discretion of the operator.

Each coil element consists of a single loop conductor with capacitors distributed in five locations around the coil conductor to minimize RF electric field losses in the patient. At gradient switching frequencies, these capacitors have a very high value of reactance to ensure that the circulating eddy currents induced in the loops from the MR system gradient coils are insignificant. At the signal output port of each anterior element, a coaxial bazooka balun, having a phase delay of 90 degrees, is used to minimize electric field losses in the coil and to minimize RF currents on the outside of the output cable shields.

As this is a receive-only coil, decoupling from the transmit RF field is needed. This is accomplished by an active decoupling circuit on each coil loop element. The decoupling network consists of a PIN diode in series with an inductor. These components are shunted across one of the five series capacitors in each coil loop adjacent to the feed capacitor. The reactance of the inductor is chosen to parallel-resonate the capacitor at the operating frequency of the coil. This places a high impedance in series with each loop at the capacitor location during transmit RF pulses, thus decoupling the coil from the transmit RF field. A shunt PIN diode in series with an inductor is used at the feedpoint of the Posterior coil element for a similar function. These diodes are biased by the system T/R driver via the coil output cable.

Each coil element is connected to the cable via an SMA connector. The system end of the external cable is terminated in a Quick Disconnect connector that mates with the Phased Array Coil Port. The Quick Disconnect box terminates in a 30 pin Bendix connector and drives two or three coil inputs - Ports 2, 5, and optionally 7. These correspond to receiver channels 0, 1 and optionally 2, respectively. The box also contains three phase compensation networks that, in combination with the length of the output cable, cause the electrical length from the PIN diodes on each coil element to the zero phase reference point of the Phased Array Coil Port to be exactly zero–an integer multiple of 1/2 wavelength, creating an effect equal to zero phase, at the operating frequency.



PHASED ARRAY SHOULDER COIL SIGNA 1.5T BLOCK DIAGRAM
ILLUSTRATION 1-1

1-2 Compatibility

The Phased Array Shoulder Coil is compatible with the following hardware configurations:

- Signa® Horizon™ 1.5T and 1.0T System (1.5T only)
- Signa® Advantage™ 1.5T, 1.0T and 0.5T System (1.5T only)
- Signa Advantage System (1.5T only)

1-3 Related Documents

- Direction 2124201-3, 5.x Signa Service Methods and 5.x/4.x/3.x Common Documents
- Direction 15400, Signa Advantage 1.5T, 1.0T, & 0.5T System
- Direction 15401, Signa Advantage 1.5T, 1.0T, & 0.5T Subsystems
- Direction 15404, Signa Advantage 1.5T, 1.0T, & 0.5T Renewal Parts
- Direction 15300, Signa Advantage System

1-4 Organization of This Document

This manual is divided into the following sections:

Section 1, Introduction - Describes Phased Array Shoulder Coil operation and when/where it can be used.

Section 2, Setup and Calibration - Describes installation procedures.

Section 3, Functional Checks - Describes the normal power-up sequence.

Section 4, Replacement/Maintenance - Describes field maintenance procedures.

Section 5, Renewal Parts - Lists field replaceable parts.

1-5 Environmental Requirements

Operate and store the Phased Array Shoulder Coil in the Scanner Room.

Dimensions: **Anterior unit -** 8.2 cm x 25.4 cm x 25.4 cm (3.25 in. x 10 in. x 10 in.)

Posterior unit - 4.0 cm x 21.9 cm x 25.7 cm (1.56 in. x 6.62 in. x 10.12 in.)

SECTION 2 - SETUP AND CALIBRATION

2-1 Checking Shipping List

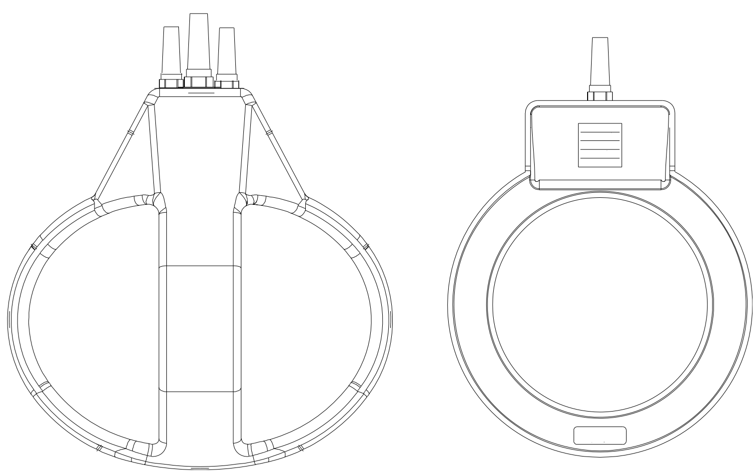
Table 2-1 lists the M1087SH Signa 1.5T Phased Array Shoulder Coil parts. Check that all parts were shipped.

TABLE 2-1
1.5T PHASED ARRAY SHOULDER COIL SHIPPING LIST

Quantity	Item	Part Number
1	1.5T Phased Array Shoulder Coil	2127695
1	Comfort Strap	E8801LD
1	Comfort Pad	E8801LN
1	Service Direction	2135786
1	Operator Manual	2131390

2-2 Installing Phased Array Shoulder Coil

1. At the Operator Console, install two new soft keys. These keys will be used by the Operator to select Shoulder imaging. Name the keys: **SHOPA2** and **SHOPA3**.



PHASED ARRAY SHOULDER COIL
ILLUSTRATION 2-1

2. Refer to the Signa, Signa Advantage and Signa Horizon System manuals for information on installing soft keys. (Use Coil/Phased Array default values in Tables 2-2 and 2-3.)

TABLE 2-2
COIL VALUES - ANTERIOR ONLY ARRAY

SYSTEM	COIL NAME	COIL TYPE	EXTREMITY COIL	CABLE LOSS	COIL LOSS	RECON SCALE FACTOR	XMIT COIL	MULTI-COIL
1.5T	SHOPA2	SURFACE	NO	1.05	0.313 $ 1.72 @	Head Coil Recon Scale Factor × 0.75	QUAD	YES

MULTICOIL NAME	NUMBER OF RECEIVERS	START RECEIVER	STOP RECEIVER	PORT ENABLE MASK	ERROR ENABLE MASK
SHOPA2	2	0	1	5	5

NUMBER OF FAST RECEIVERS	START FAST RECEIVER	STOP FAST RECEIVER	TRANSMIT ATTENUATION
0	4	4	0

Note: $ - 55 cm
@ - 60 cm

TABLE 2-3
COIL VALUES - ANTERIOR & POSTERIOR ARRAY

SYSTEM	COIL NAME	COIL TYPE	EXTREMITY COIL	CABLE LOSS	COIL LOSS	RECON SCALE FACTOR	XMIT COIL	MULTI-COIL
1.5T	SHOPA3	SURFACE	NO	1.05	0.313 $ 1.72 @	Head Coil Recon Scale Factor × 0.75	QUAD	YES

MULTICOIL NAME	NUMBER OF RECEIVERS	START RECEIVER	STOP RECEIVER	PORT ENABLE MASK	ERROR ENABLE MASK
SHOPA3	4	0	3	13	13

NUMBER OF FAST RECEIVERS	START FAST RECEIVER	STOP FAST RECEIVER	TRANSMIT ATTENUATION
0	4	4	0

Note: $ - 55 cm
@ - 60 cm

2-3 Functional Checks

1. Perform a Body coil scan performance verification. Refer to *Section 3-1, Signa Advantage 1.5T Scanner Verification*.

2. Perform a Phased Array Shoulder Coil performance Verification. Refer to *Section 3-2, Phased Array Shoulder Coil Imaging Performance Verification*.

3. Perform an SNR Image analysis. Refer to *Section 3-3-1, SNR Image Analysis for 4.x Systems* or *Section 3-3-2, SNR Image Analysis for 5.x Systems*.

4. Record the Original SNR value in the SNR QA Data Table in *Section 6, Forms*.

2-4 Periodic Quality Assurance Check

On a periodic basis, such as during planned maintenance, perform the quality assurance checks outlined below to ensure that the coil is operating properly.

1. Check the external cable for cracks or breaks once a week. Refer to *Section 4-6, Checking External Cable*.

2. Perform a coil SNR verification. Refer to *Section 3-2, Phased Array Shoulder Coil Imaging Performance Verification*.

3. Record the date and value calculated in *Section 3-3, SNR Image Analysis* in Column 2 under "SNR Data QA Check" of the Data Table (found in *Section 6, Forms*) as instructed.

4. As instructed in the Data Table, divide the SNR value obtained in the periodic QA Check by the original SNR value and record in Column 3 of the Data Table.

5. If this ratio is **not** greater than 85%, there may be a problem in the coil system. Contact your local GE Service Representative.

SECTION 3 - FUNCTIONAL CHECKS

3-1 Sigan Advantage 1.5T Scanner Verification

Note:
An alternate proprietary procedure is available for GE use and to customers with a valid Advanced Service Package Limited License. Refer to *System Level Procedures: Troubleshooting: TLT Procedure.*

Phantom Required:

- Head TLT Phantom, P/N 46-265826G6
- Universal Phantom Holder, P/N 46-328383P1

Setup Procedure:

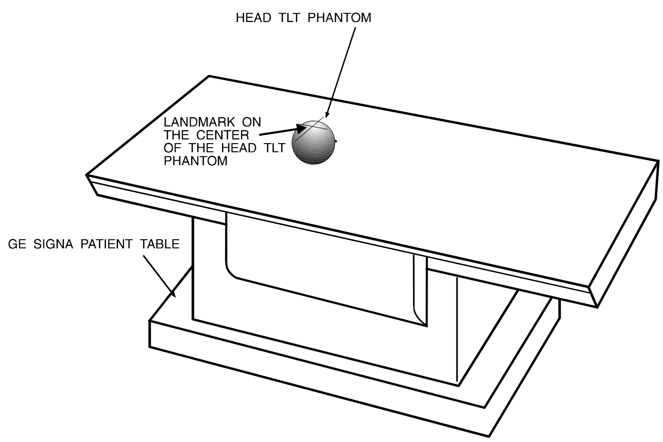
CAUTION

The Quad Head Coil must be completely removed from the cradle before performing any body scans. Failure to do this may result in damage to the Quad Head Coil T/R Network.

1. Remove Quad Head Coil (if present) from the cradle.

2. Select **[NEW STUDY] (4.x)** or **[New Exam] (5.x)** to allow a new Landmark to be set.

3. Position the Head TLT phantom in the center of the cradle. (The Universal Phantom Holder can be used to keep the TLT phantom stable.) Landmark the center of the phantom and advance to isocenter using the **[ADVANCE TO SCAN]** button. (See Illustration 3-1.)



HEAD PHANTOM LANDMARK SETUP
ILLUSTRATION 3-1

4a. **For 4.x:** Set up scan prescription as shown in Table 3-1.

TABLE 3-1
BODY COIL SNR - SCAN PROTOCOL (4.X)

ID:	**geservice**	Echo Time (TE):	**[40 msec]**
Name:	**body snr**	Rep Time (TR):	**[400 msec]**
Patient Weight:	**111**	Auto CF:	**[Peak]**
Patient Entry:	**[Head First]**	Field of View:	**[32 cm]**
Patient Position:	**[Supine]**	Scan Thickness:	**[3 mm]**
Axial/Sag Landmark:	**[Sternal Notch]**	Scan Location:	**(S/I) 0**
Coil Type:	**[Body Coil]**	FOV Center:	**(R/L) R0 (A/P) A0**
Scan Plane:	**[Axial]**	Acq. Matrix:	**[256 x 256]**
Image Mode:	**[Single Scan]**	Frequency Direction:	**[R/L]**
Pulse Sequence:	**[Multiple Echo]**	Imaging Time:	**[1 NEX 1:48]**
Imaging Options:	**[None]**	Contrast:	**[No]**
or enter PSD filename		Table Delta:	**0 mm**
Number of Echoes:	**[1]**		

4b. **For 5.x:** Set up scan prescription as shown in Table 3-2.

TABLE 3-2
BODY COIL SNR - SCAN PROTOCOL (5.X)

ID:	**geservice**	Rep Time (TR):	**[400 msec]**
Name:	**body snr**	Auto CF:	**[Peak]**
Patient Weight:	**111**	Field of View:	**[32 cm]**
Patient Entry:	**[Head First]**	Scan Thickness:	**[3 mm]**
Patient Position:	**[Supine]**	Interscan Spacing:	**[Other] 0**
Axial/Sag Landmark:	**[Sternal Notch]**	Start Loc (I/S): **0**	End Loc (I/S): **0**
Coil Type:	**[Body Coil]**	No. of Scan Location:	**1**
Scan Plane:	**[Axial]**	FOV Center (L/R): **0**	(A/P): **0**
Image Mode:	**[2D]**	Acq. Matrix (freq.):	**[256]**
Pulse Sequence:	**[Spin Echo]**	Acq. Matrix (phase):	**[256]**
Imaging Options:	**[None]**	Frequency Direction:	**[A/P]**
or enter PSD filename		Imaging Time:	**[1 NEX 1:49]**
Number of Echoes:	**[1]**	Contrast:	**[No]**
Echo Time (TE):	**[20 msec]**	Table Delta:	**0 mm**

5. Select **[Auto Prescan]** to properly calibrate the RF power level for the 90-degree and 180-degree pulses.

6. Select **[Scan]**, and observe the resulting images. Ensure that there are no artifacts of any sort in the resulting image. Record the Exam number and Series number for SNR calculations.

7. Select **[Scan]** again. This second image will be used for determination of Body Coil mode SNR.

8. Select **[Cancel]**. Refer to *Section 3-3, SNR Image Analysis*.

3-2 Phased Array Shoulder Coil Imaging Performance Verification

Note:
An alternate proprietary procedure is available for GE use and to customers with a valid Advanced Service Package Limited License. Refer to *System Level Procedures: Troubleshooting: TLT Procedures.*

Phantom Required:

Head TLT Phantom, P/N 46-265826G6

Setup Procedure:

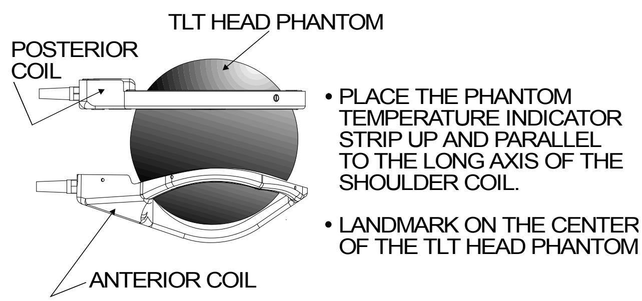
1. Select **[NEW STUDY] (4.x)** or **[New Exam] (5.x)** to allow a new Landmark to be set.

CAUTION

> **The Quad Head Coil must be completely removed from the cradle before performing any body scans. Failure to do this may result in damage to the Quad Head Coil T/R Network.**

2. Remove Quad Head Coil (if present) from the cradle.

3. Place the Anterior Coil of the Phased Array Shoulder Coil to be tested upside down on the patient table. Use a wedge pad to support the coil. Place the TLT Head Phantom on the coil and the Posterior Coil on the top of the phantom. (See Illustration 3-2.)

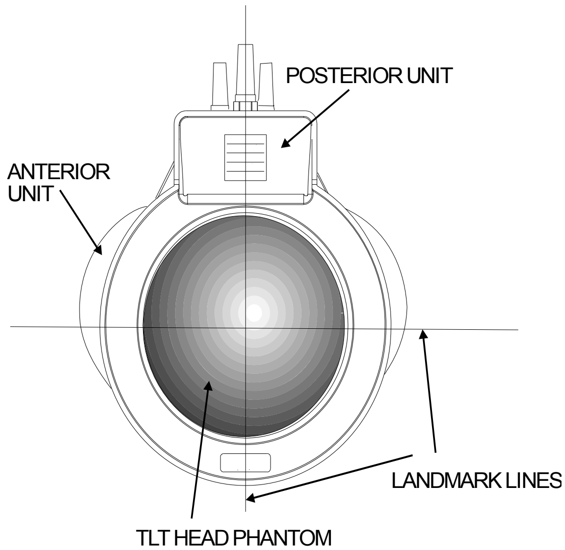


PHASED ARRAY SHOULDER/TLT HEAD PHANTOM SETUP
ILLUSTRATION 3-2

4. Attach the Velcro® straps around the external cables to secure the two cables close to each other.

5. Connect the Phased Array Shoulder Coil connector to the mating connector on the Carriage Assembly, then connect the Posterior Coil to the Phased Array Shoulder Coil Connector.

6. Position the Phased Array Shoulder Coil in the center of the table. Landmark for center of TLT Head Phantom, and advance to isocenter using the **[ADVANCE TO SCAN]** button. (See Illustration 3-3.)



PHASED ARRAY SHOULDER LANDMARK
ILLUSTRATION 3-3

7a. **For 4.x:** Set up scan prescription as shown in Table 3-3.

TABLE 3-3
PHASED ARRAY SHOULDER COIL SNR - SCAN PROTOCOL (4.X)

ID:	**geservice**	Echo Time (TE):	**[40 msec]**
Name:	**pa sholdr snr**	Rep Time (TR):	**[400 msec]**
Patient Weight:	**111**	Auto CF:	**[Peak]**
Patient Entry:	**[Head First]**	Field of View:	**[32 cm]**
Patient Position:	**[Supine]**	Scan Thickness:	**[3 mm]**
Axial/Sag Landmark:	**[Sternal Notch]**	Scan Location:	**(S/I) 0**
Coil Type:	**[Other Coil] [shopa3]**	FOV Center:	**(R/L) R0 (A/P) A0**
Scan Plane:	**[Axial]**	Acq. Matrix:	**[256 x 256]**
Image Mode:	**[Single Scan]**	Frequency Direction:	**[A/P]**
Pulse Sequence:	**[Multiple Echo]**	Imaging Time:	**[1 NEX 1:47]**
Imaging Options:	**[None]**	Contrast:	**[No]**
or enter PSD filename		Table Delta:	**0 mm**
Number of Echoes:	**[1]**		

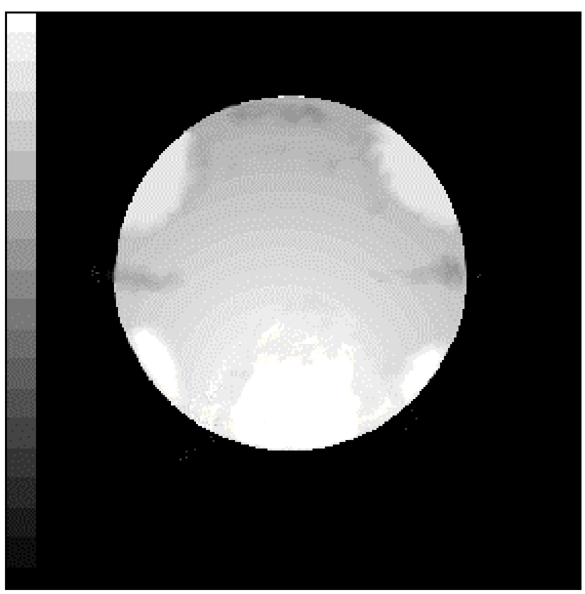
7b. **For 5.x:** Set up scan prescription as shown in Table 3-4.

TABLE 3-4
PHASED ARRAY SHOULDER COIL SNR - SCAN PROTOCOL (5.X)

ID:	**geservice**	Rep Time (TR):	**[400 msec]**
Name:	**pa sholdr snr**	Auto CF:	**[Peak]**
Patient Weight:	**111**	Field of View:	**[32 cm]**
Patient Entry:	**[Head First]**	Scan Thickness:	**[3 mm]**
Patient Position:	**[Supine]**	Interscan Spacing:	**[Other] 0**
Axial/Sag Landmark:	**[Sternal Notch]**	Scan Loc (I/S): **0**	End Loc (S/I): **0**
Coil Type:	**[Other Coils][shopa3]**	No. of Scan Location:	**1**
Scan Plane:	**[Axial]**	FOV Center (L/R):	**0** (A/P) **0**
Image Mode:	**[2D]**	Acq. Matrix (freq.):	**[256]**
Pulse Sequence:	**[Spin Echo]**	Acq. Matrix (phase):	**[256]**
Imaging Options:	**[None]**	Frequency Direction:	**[R/L]**
or enter PSD filename		Imaging Time:	**[1 NEX 1:47]**
Number of Echoes:	**[1]**	Contrast:	**[No]**
Echo Time (TE):	**[40 msec]**	Table Delta:	**0 mm**

8. Select **[Auto Prescan]** to properly calibrate the RF power level for the 90-degree and 180-degree pulses.

9. Select **[Scan]**, and observe the resulting image of the sphere. (Illustration 3-4 shows a normal image.) Ensure that there are no artifacts of any sort in the sphere image. Record the Exam number and Series number for SNR Calculations.



PHASED ARRAY SHOULDER COIL IMAGE
ILLUSTRATION 3-4

10. Select **[Scan]** again. This second image of the sphere will be used for determination of Phased Array Shoulder mode SNR.

11. Select **[Cancel]**. Refer to Section 3-3 for SNR Image Analysis.

3-3 SNR Image Analysis

3-3-1 SNR Image Analysis (Release 4.x)

See Section 3-3-2 for Release 5.x.

Description:

The CLIPS macro will ask for the first image in the series to analyze. The macro assumes two back-to-back scans were done, and will take the designated image and the next image in the series (i.e., if image 3 is provided, it will take 3 and 4). It will also ask that three points on the border of the phantom image be designated to use in positioning the analysis ROI. If the image is too big or small, or the three points were selected too close together, the macro will inform you.

Procedure:

1. Touch **[UTILITIES]**, then **[Clips]**.

```
                  CLIPS
          -------------------------
              1 Run CLIPS
              2 Save files
              3 Restore files
              4 Remove files
              5 Exit
```

`Enter the number of your selection: ..........................` **1 [ENTER]**

```
        1 = Operator console
        2 = Remote1
        3 = Remote2
        4 = Exit
```

`Which image processor would you like to use (1 2 or 3)?.......` **1 [ENTER]**

`IP selected is at the operator console.`

`Do you wish to boot the Image Processor? (Y or N) ............`**Y [ENTER]**

Note: It is not necessary to reboot the IP each time you enter.

`Welcome to the Command Line Image Processing System (CLIPS)`

`CLIPS > ...............................................`**LIST(STUDY); [ENTER]**

```
                              STUDY LIST
=========================================================================
   STUDY #         PATIENT NAME        PATIENT I.D.        DATE
-------------------------------------------------------------------------
    00001            BDY_SNR             GESERVICE       9-JULY-89
    00002            HD_SNR              GESERVICE       9-JULY-89
-------------------------------------------------------------------------

CURRENT DEFAULTS --- PATIENT ID =                    /  /

                      N - NEXT PAGE
                      P - PREVIOUS PAGE
                      S - SELECT STUDY
                      C - CANCEL
```

choice ?...**S [ENTER]**
ENTER STUDY NUMBER ....**1 [ENTER]**

Note: Enter appropriate number.

CLIPS > ....**LIST(SERIES); [ENTER]**

```
                             SERIES LIST
=========================================================================
 SERIES #         # OF IMAGES            SERIES TYPE        PULSE TYPE
-------------------------------------------------------------------------
   001              002                     AXIAL              ME
   002              002                     SAG                ME
-------------------------------------------------------------------------
CURRENT DEFAULTS --- PATIENT ID = SNR            0000S/00S/001
                             N - NEXT PAGE
                             P - PREVIOUS PAGE
                             S - SELECT SERIES
                             C - CANCEL
```

Choice ? ....**S [ENTER]**
ENTER SERIES NUMBER --...**1 [ENTER]**

Note: Enter appropriate number.

CLIPS > ....**DIS; [ENTER]**

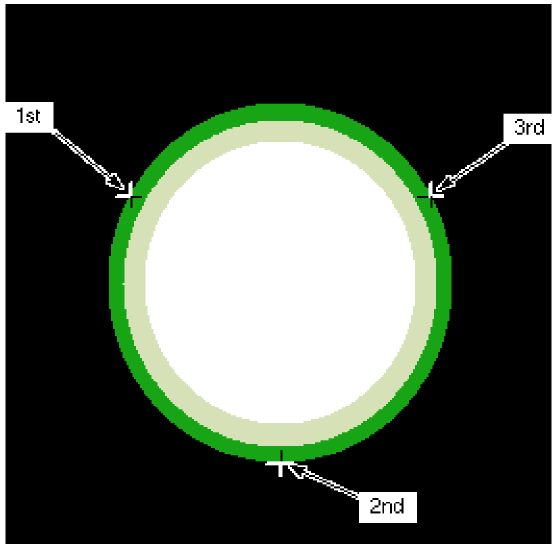
or: LIST(IMAGE) to select the image desired.

CLIPS > ....**EXECUTE(SNR); [ENTER]** or **EXE(SNR); [ENTER]**

```
THIS MACRO CALCULATES THE SNR USING THE GE HEAD OR BODY PHANTOM.
ENTER THE NUMBER OF THE FIRST IMAGE OF THE TWO-IMAGE SET AND !
EXAMPLE INPUT> 1!
Please start entering data.  Type an ! to end input mode.
```

CLIPS INPUT> ....**1! [ENTER]**

Correct positioning of the SNR analysis ROI depends on finding the center of the phantom. You will be prompted to identify three points at the edge of the phantom. Care should be taken to pick three points as far away from each other as possible, i.e., 2, 6 and 10 o'clock, and avoid flat spots and irregularities on the edge. (See Illustration 3-5.)



SNR CURSOR POSITIONS
ILLUSTRATION 3-5

Note:
The order of cursor placement must be 10, 6, then 2 o'clock positions. Using a different order may give an error.

```
POSITION CURSOR AT FIRST POINT AND ENTER !
```

Note: Place cursor at 10:00 position on edge of image.

```
Please start entering data.  Type an ! to end input mode.
CLIPS INPUT> ...............................................! [ENTER]
```

```
POSITION CURSOR AT SECOND POINT AND ENTER !
```

Note: Place cursor at 6:00 position.

```
Please start entering data.  Type an ! to end input mode.
CLIPS INPUT> ...............................................! [ENTER]
```

```
POSITION CURSOR AT THIRD POINT AND ENTER !
```

Note: Place cursor at 2:00 position.

```
Please start entering data.  Type an ! to end input mode.
CLIPS INPUT> ...............................................! [ENTER]
```

```
PHANTOM POINTS AND RADIUS OK.  BEGIN ANALYSIS.
```

Note: The program will perform the analysis and display the final values on the image monitor. Record Signal, Noise and Signal-to-Noise in the Data Table in Section 6.

```
DO YOU WANT TO SAVE THIS DATA?
ENTER A  1! FOR YES, A  0! FOR NO.
Please start entering data.  Type an ! to end input mode.
CLIPS INPUT>.................................................. 0! [ENTER]

DATA NOT SAVED
CLIPS >  .........................................LIST(SERIES); [ENTER]
```

Note: Select the next series and repeat the above analysis for each image pair. Type **BYE** when completed to exit CLIPS.

2. When all analyses are complete, reset Auto Center Frequency mode by touching **[SCAN MODES]**, **[Default Auto CF]**, then **[EXECUTE]**.

   Note:
   It is not necessary to reset the Auto Center Frequency mode if proceeding onto more image quality scans.

3. Record the date and value calculated in the appropriate column under "SNR Data QA Check" of the Data Table (found in *Section 6, Forms*) as instructed.

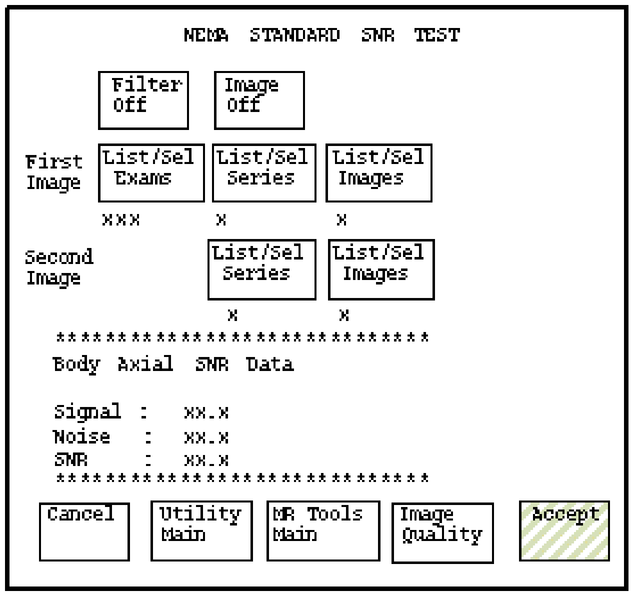
3-3-2 SNR Image Analysis (Release 5.x)

Description:

The SNR tool retrieves two operator-selected images. Signal value is computed as the mean pixel value in a ROI covering 80% of the image. The image is analyzed to determine the center of the image for positioning the ROI. A difference image is created by subtracting the second image from the first, and the same ROI is used to calculate noise from the subtracted image. The Signal value, Noise value, and Signal-to-Noise ratio are reported. There is an option to save the difference image with the results annotated.

Procedure:

1. Touch **[UTILITIES]**, **[MR Tools]**, **[Image Quality]**, then **[SNR Test]**. The SNR Test screen in Illustration 3-6 displays.



Note: Accept changes to continue only after an analysis has been performed.

SNR TEST SCREEN
ILLUSTRATION 3-6

2. Enter image exam, series, and image numbers. If exam, series, or image numbers are not known, select **[List Exams]**, **[List Series]**, or **[List Images]** to display list from which to choose.

   Note:
   Image number selection must be backlit (highlighted) to be able to enter information. Use the Switch key on the keyboard to transfer control from the left to right side of touch screen.

3. If high pass filtering is desired to be performed on data, touch **[Filter Off]**, which will highlight and change to **[Filter On]**.

4. If the difference image annotated with data is to be created, touch **[Image Off]**, which will highlight and change to **[Image On]**.

5. Touch **[Accept]** to begin analysis. The final values are displayed on the touch screen. (See Illustration 3-6, which shows the previous screen.)

6. Touch **[Continue]**, select the next exam, and repeat the above analysis for each image pair.

7. Record the date and value calculated in the appropriate column under "SNR Data QA Check" of the Data Table (found in *Section 6, Forms*) as instructed.

3-4 Checking Anterior Coil External Cable and PIN Diodes

Note:
The Posterior Coil must be plugged in for the measurements that follow.

1. Select the **DIODE TEST** function on the Digital Multimeter (DMM). Ensure that the Posterior Coil is connected.

2. Connect the positive and negative leads as detailed in Table 3-5. Verify that a reading of 0.400 to 0.900 is obtained at each test point.

TABLE 3-5
DIODE TEST CONNECTIONS

Coil Type	Positive Lead Connection		Negative Lead Connection	
	For Step 2	For Step 5	For Step 2	For Step 5
A1	3B	3A	3A	3B
A2	9B	9A	9A	9B
P	13B	13A	13A	13B

- If a reading of less than 0.400 is obtained, a PIN diode or the coil output cable is likely to be shorted.

- If a reading of more than 0.900 is obtained, a PIN diode or the coil output cable is likely to be open.

3. Connect the positive and negative leads as shown in Table 3-5. Verfiy that a reading of INFINITY (overrange) is obtained at each test point.

   If a finite reading is obtained, a PIN diode is likely to be shorted or leaky, or the coil output cable is likely to be shorted.

Important:
Due to safety hazards of soldering in the field, PIN diodes are no longer FRUs. If the coil has a defective PIN diode, the respective coil component must be replaced.

3-5 Checking Posterior Coil Cable and PIN Diode

Note:
Disconnect the Posterior Coil cable from the Anterior Coil Connector.

1. Select the **DIODE TEST** function on the Digital Multimeter (DMM).

2. Connect the positive and negative leads as described in TABLE 3-6 directly to the Posterior element coaxial fitting. Verify that a reading of 0.400 to 0.900 is obtained at the test point.

TABLE 3-6
POSTERIOR PADDLE CABLE TEST CONNECTIONS

Coil Type	Positive Lead Connection		Negative Lead Connections	
	For Step 2	For Step 5	For Step 2	For Step 5
P	J4 center	J4 shell	J4 shell	J4 center

- If a reading less than 0.400 is obtained, the PIN diode or cable is likely to be shorted.
- If a reading of more than 0.900 is obtained, the PIN diode or cable is likely to be open.

3. Connect the positive and negative leads as shown in Table 3-6. Verify that a reading of INFINITY (overrange) is obtained.

   If a finite reading is obtained the PIN diode is likely to be shorted to leaky, or the posterior paddle cable is defective.

3-6 Checking PIN Diodes at Coil - Anterior and Posterior

Disassemble the coil as described in Section 4-1, and remove the output cable of the Phased Array Shoulder Coil.

1. Select the **DIODE TEST** function on the Digital Multimeter (DMM).
2. Connect the positive and negative leads as described in TABLE 3-7 directly to the coil element coaxial fitting. Verify that a reading of 0.400 to 0.900 is obtained at the test point.

TABLE 3-7
DIODE TEST CONNECTIONS

Coil Type	Positive Lead Connection		Negative Lead Connection	
	For Step 2	For Step 5	For Step 2	For Step 5
A1	J1 center	J1 shell	J1 shell	J1 center
A2	J2 center	J2 shell	J2 shell	J2 center
P	J3 center	J3 shell	J3 shell	J3 center

- If a reading less than 0.400 is obtained, the PIN diode is likely to be shorted.

- If a reading of more than 0.900 is obtained, the PIN diode is likely to be open.

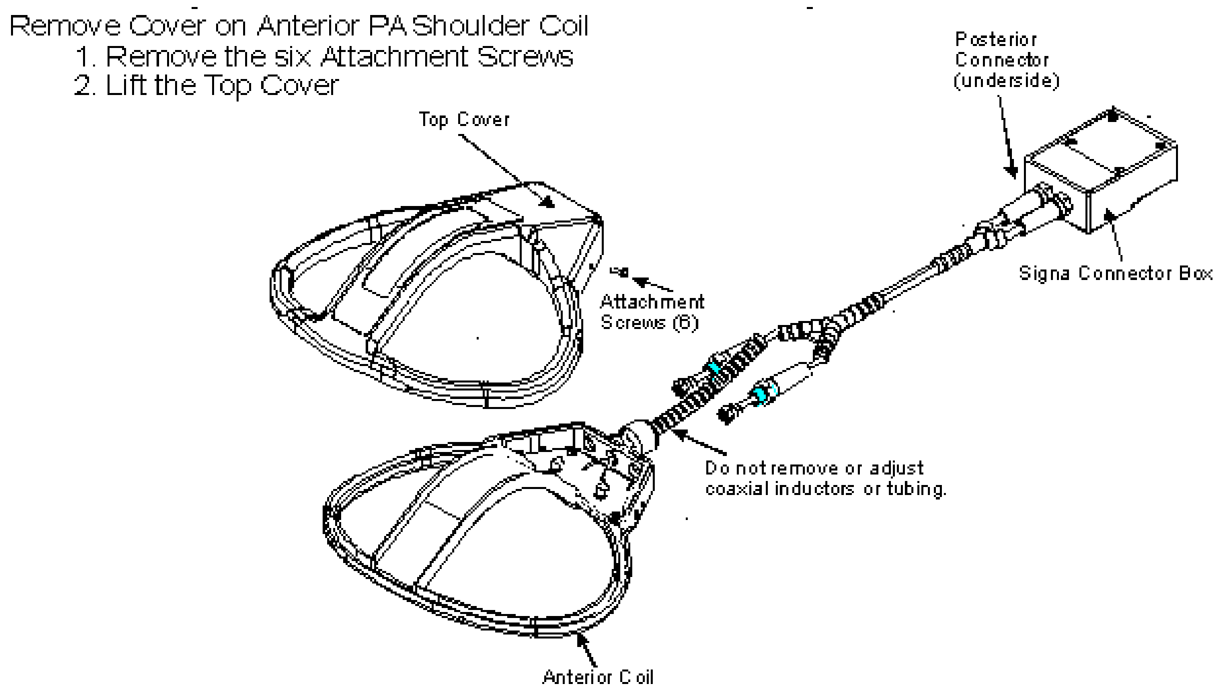
3. Connect the positive and negative leads as shown in TABLE 3-7. Verify that a reading of INFINITY (overrange) is obtained.

   If a finite reading is obtained the PIN diode is likely to be shorted or leaky.

   Important:
   Due to safety hazards of soldering in the field, PIN diodes are no longer FRUs. If the coil has a defective PIN diode, the respective coil component must be replaced.

SECTION 4 - REPLACEMENT / MAINTENANCE

4-1 Disassembly of Anterior Coil of Phased Array Shoulder Coil

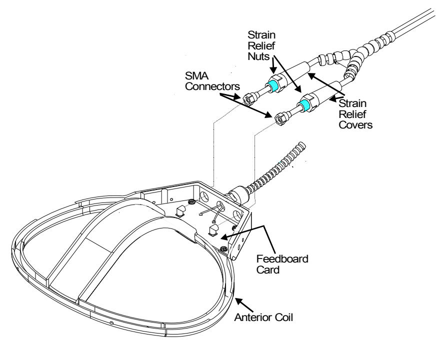


REPLACEMENT / MAINTENANCE
ILLUSTRATION 4-1

4-2 Replacing External Cable

External Coil Cable Removal

1. Unscrew the Strain Relief Covers and pull away from the Strain Relief Nuts.
2. Disconnect the SMA Connectors from the Feedboard Card.
3. Unscrew the Strain Relief Nuts.
4. Back the SMA Connectors through the threaded holes.



External Coil Cable Replacement

1. Insert the SMA Connectors through the threaded holes.
2. Connect the SMA Connectors in the same way as the original.
3. Insert and tighten the Strain Relief Nuts. Tighten the Strain Relief Covers.

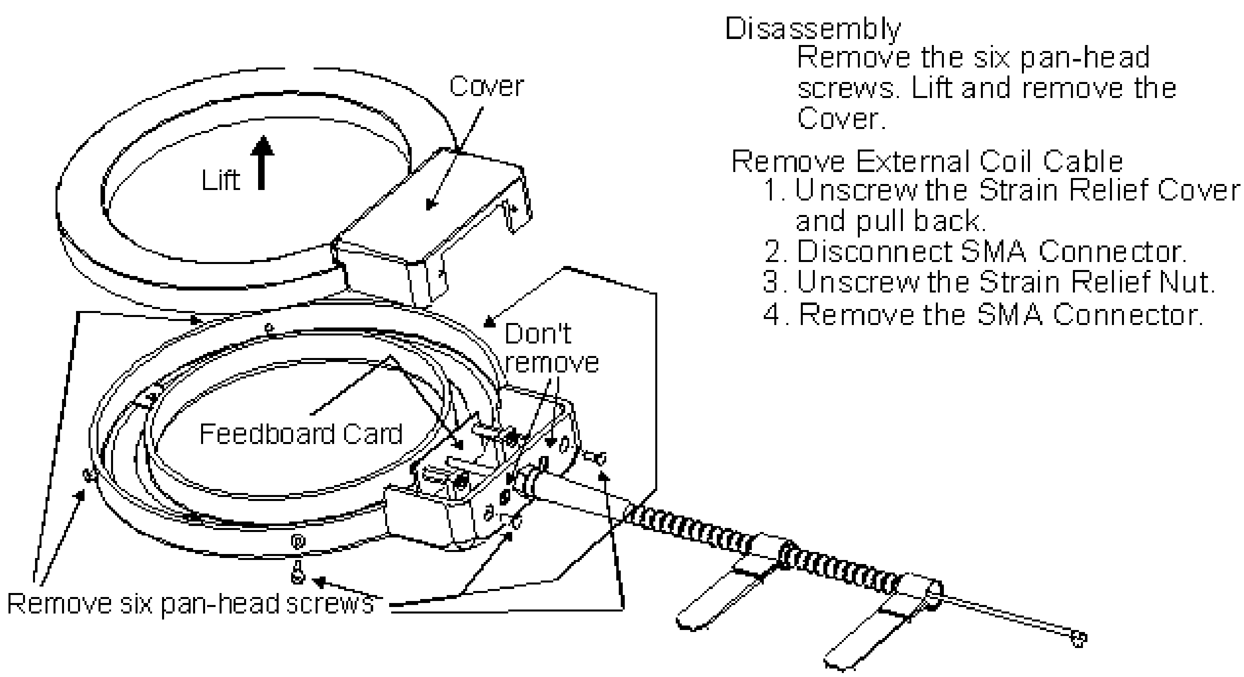
Anterior Coil Assembly

1. Replace cover, insert and tighten screws.

EXTERNAL CABLE REMOVAL, REPLACEMENT, AND ASSEMBLY
ILLUSTRATION 4-2

Repeat the performance tests in Section 3-2.

4-3 Disassembly of Posterior Coil and Removing External Cable



DISASSEMBLY OF POSTERIOR COIL AND REMOVING EXTERNAL CABLE
ILLUSTRATION 4-3

4-4 Replacing Mechanical Hardware

Replacing External Cable:

1. Insert the SMA Connector through the threaded hole.
2. Connect the SMA Connector. The quick disconnect end should point away from the open coil.
3. Insert and tighten the Strain Relief Nut, and tighten the Strain Relief Cover.

Assembly of Posterior Coil:

1. Replace the cover of the coil.
2. Insert and tighten the six pan-head screws.
3. Repeat the Performance Tests in Section 3-2.

4-5 Replacing Mechanical Hardware (Except Housing Screws)

Replace the entire Phased Array Shoulder Coil. Refer to *Section 5, Renewal Parts* for the Phased Array Shoulder Coil part number.

4-6 Checking External Cable

Check the external cables for cracks or breaks once each week. Replace the appropriate Phased Array Shoulder Coil cable if any damage or wear is found on the external cables.

4-7 Cleaning Coil

CAUTION

> **Avoid damaging sensitive electronic parts. Do not spray or pour dishwashing solution directly onto the Phased Array Shoulder Coil, or external cable. Do not use alcohol to clean the Phased Array Shoulder Coil. Never submerge the Phased Array Shoulder Coil in any liquid.**

1. Clean the Phased Array Shoulder Coil and external cable with a mild dishwashing liquid and water solution. Wet a soft cloth with the solution and proceed to clean.
2. Before returning the coil for servicing, thoroughly clean the coil, cable, and connectors with a 10% bleach in water solution.

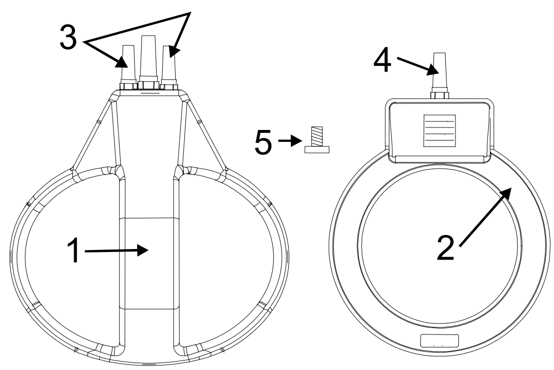
5- RENEWAL PARTS

WARNING!

BIOHAZARD! EQUIPMENT BEING RETURNED FROM USE IN A CLINICAL SETTING MUST BE CLEAN AND FREE OF BLOOD AND OTHER INFECTIOUS SUBSTANCES. THE DEPARTMENT OF TRANSPORTATION (DOT) HAS RULED THAT ITEMS THAT WERE SATURATED AND/OR DRIPPING WITH HUMAN BLOOD THAT ARE NOW CAKED WITH DRIED BLOOD; OR WERE USED OR INTENDED FOR USE IN PATIENT CARE, ARE REGULATED MEDICAL WASTE FOR TRANSPORTATION PURPOSES AND MUST BE TRANSPORTED AS A HAZARDOUS MATERIAL. UNDER NO CIRCUMSTANCES SHOULD A PART OR EQUIPMENT WITH VISIBLE BODY FLUIDS BE TAKEN OR SHIPPED FROM A CLINIC OR SITE (FOR EXAMPLE, SURFACE COILS).

Employees shall follow proper decontamination procedures for clean up of blood borne pathogens. Refer to *Section 4-7, Cleaning Coil*. It is the responsibility of the GEMS employee to ensure the part/equipment is properly decontaminated prior to shipment.

1.5T Phase Array Shoulder Coil, P/N 2131390



Item	Part Number	FRU	Name	Quantity	Description (Remarks)
1	2100937-17	1	MEDRAD FRU	1	305002514 1.5T PA ANTERIOR w/CABLE
2	2100937-19	1	MEDRAD FRU	1	305002516 1.5T PA POSTERIOR w/CABLE
3	2100937-18	1	MEDRAD FRU	1	305002515 1.5T PA ANTERIOR w/CABLE
4	2100937-20	1	MEDRAD FRU	1	305002517 1.5T PA POSTERIOR w/CABLE
5	46-208922P4	2	SCREW, MACH	12	6-32 x 0.375 NYLON SCREWS

SECTION 6 – FORMS

DATA TABLE

Use the space provided below to record the calculated Signal to Noise Ratio (SNR) data obtained from *Section 3, Functional Checks*.

After recording the SNR data obtained during the initial coil installation, record subsequent SNR data in Column 2 below and the date they were obtained in Column 1 as a periodic QA check. If the ratio of any of the coils found is **not** greater than 85% (Column 3), there is a problem in the coil or the MR system.

Original SNR data obtained at initial coil installation:

Body Coil SNR Value:_________________

Phased Array Shoulder Coil SNR Value: Both (Anterior/Posterior)____________

Date: ______________

SNR Data QA (Quality Assurance) Check

1	2	3
Date	Posterior/Anterior	Are new Values Divided by Original Values > 85%?
__________	__________	__________ Y/N
__________	__________	__________ Y/N
__________	__________	__________ Y/N
__________	__________	__________ Y/N
__________	__________	__________ Y/N
__________	__________	__________ Y/N
__________	__________	__________ Y/N
__________	__________	__________ Y/N
__________	__________	__________ Y/N
__________	__________	__________ Y/N
__________	__________	__________ Y/N

Note: Make additional copies of this document as required.

DEFECTIVE COIL RETURN FORM

Note:
To allow for proper assessment of defective returned coils, this form must be completely filled out and accompany all returned coils. Include films or prints of any image quality related complaints with a description of scan protocol used.

Date ____

Site Name ____

Site Address ______________________________________

Service Engineer ___________________________________

Coil Serial Number__________________________________

Date Coil Installed __________________________________

Description of Coil Problem ___________________________

____

____

____

____

____

ELECTRICAL CHECKS

PIN Diode Test

Meter Reading Forward Bias – Anterior D1 ________________________

Meter Reading Reverse Bias – Anterior D1 ________________________

Meter Reading Forward Bias – Anterior D2 ________________________

Meter Reading Reverse Bias – Anterior D2 ________________________

Meter Reading Forward Bias – Posterior D1 ________________________

Meter Reading Reverse Bias – Posterior D1 ________________________

Meter Reading Forward Bias – Posterior D2 ________________________

Meter Reading Reverse Bias – Posterior D2 ________________________

Note:
An alternate proprietary procedure is available for GE use and to customers with a valid Advanced Service Package Limited License. Refer to *System Level Procedures: Troubleshooting: TLT Procedures.* Use Table DR-1 and DR-2 to record the TLT results of the defective coil.

TABLE DR-1
TLT DATA FOR DEFECTIVE SURFACE COIL

SITE: ________________________________ **NAME:** ____________________________

DATE: ___________________________

TLT FILE NUMBER: ___________________________

SNR: NOISE ___________________________

SNR MEAN ___________________________

SNR SIGNAL ___________________________

SNR AREA ___________________________

TR MAP: 89-91 ___________________________ %

85-95 ___________________________ %

65-115 ___________________________ %

FLIP MEAN ___________________________

TABLE DR-2
TLT DATA FOR DEFECTIVE PHASED ARRAY COIL

SITE: ______________________________ **NAME:** ____________________________

DATE: _______________________

TLT FILE NUMBER: __________________________

SNR: NOISE (0) __________________________

NOISE (1) __________________________

NOISE (2) __________________________

NOISE (3) __________________________

NOISE (C) __________________________

SNR MEAN (0) __________________________

SNR MEAN (1) __________________________

SNR MEAN (2) __________________________

SNR MEAN (3) __________________________

SNR MEAN (C) __________________________

TR MAP: 89-91 (C) __________________________%

(Combined 85-95 (C) __________________________%

Results) 65-115 (C) __________________________%

FLIP MEAN (C) __________________________

FLIP SDV (C) __________________________

RCV MEAN (C) __________________________

RCV SDV (C) __________________________

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